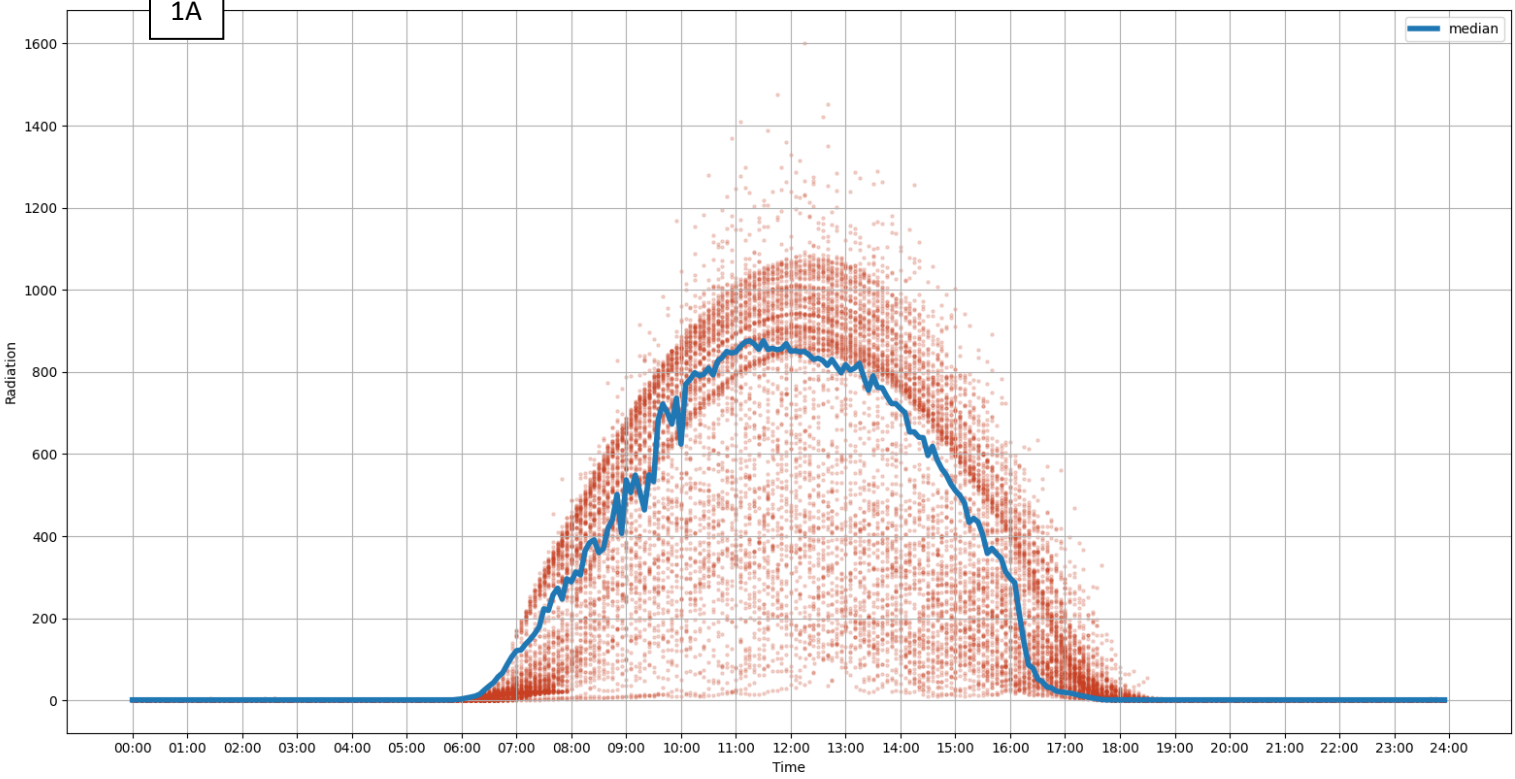
PS:2 Data Analysis Report

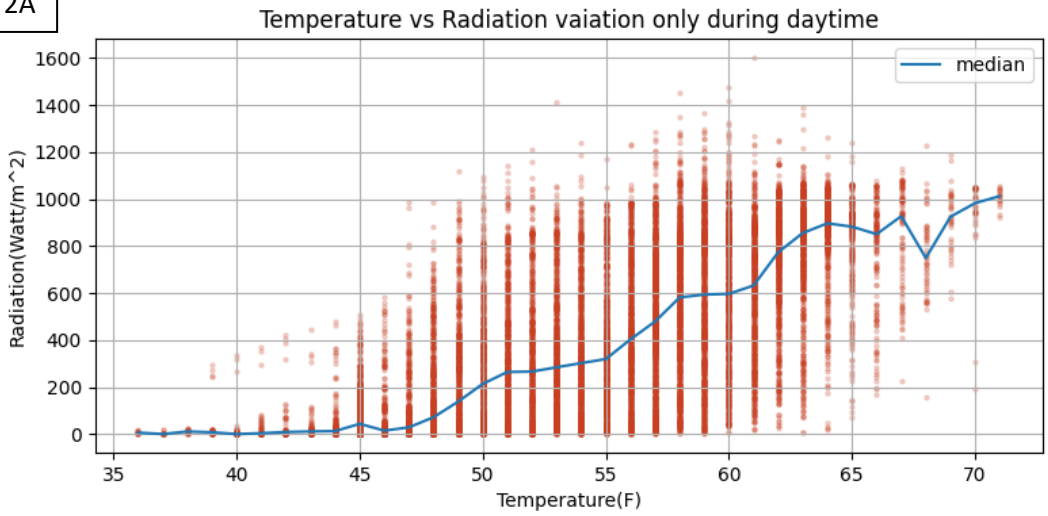
A. Data visualization

1A

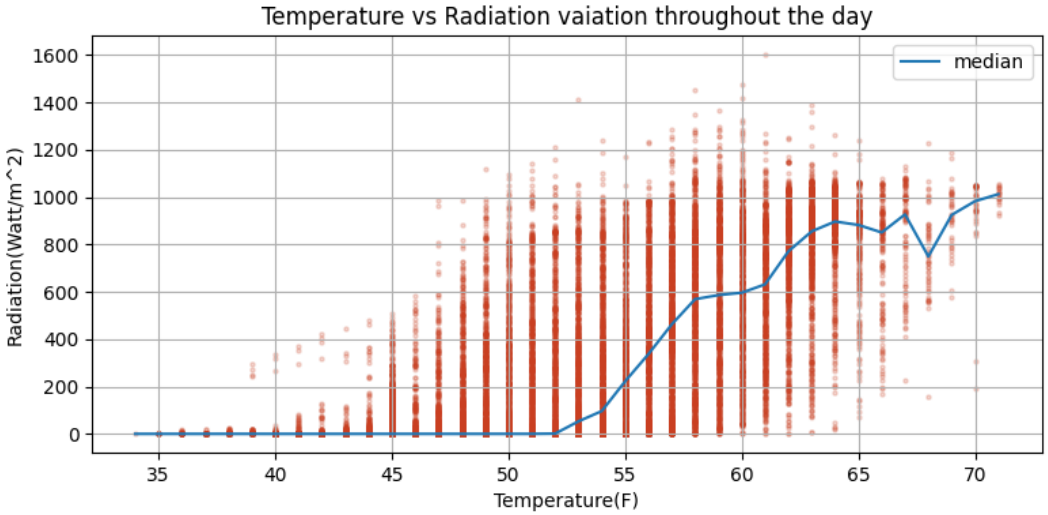


The above graph shows the variation of the radiation received at different times of the day. The red points are the values of radiation measured in watt/m^2 plotted as observed at a given time of the day for the different dates. The blue line represents the median value (across the different days) of the recorded radiation at the different time of the day.

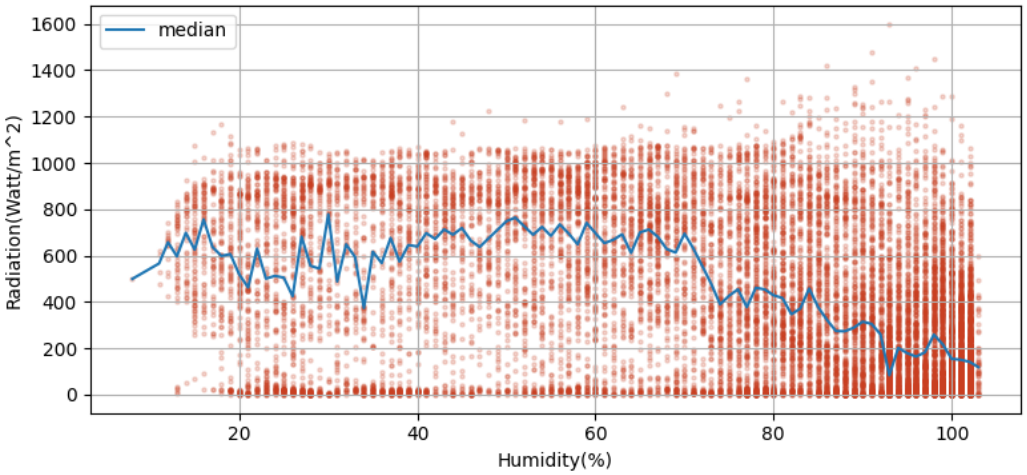
2A


Temperature vs Radiation vaiation only during daytime
median
Radiation(Watt/m^2)
Temperature(F)
0
200
400
600
800
1000
1200
1400
1600
35
40
45
50
55
60
65
70

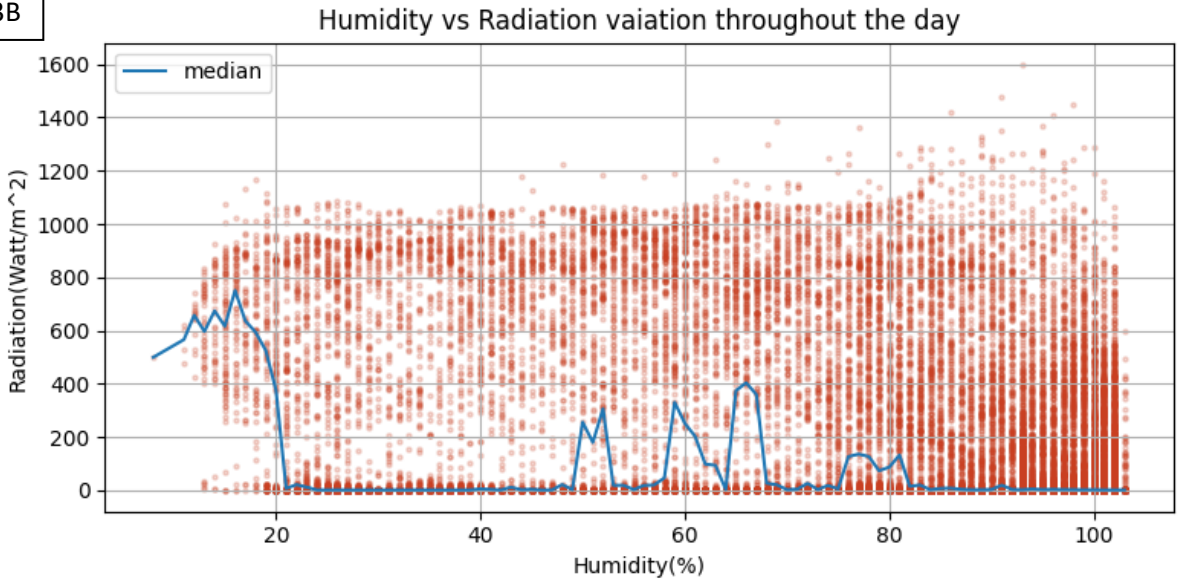
2B


Temperature vs Radiation vaiation throughout the day
median
Radiation(Watt/m^2)
Temperature(F)
0
200
400
600
800
1000
1200
1400
1600
35
40
45
50
55
60
65
70

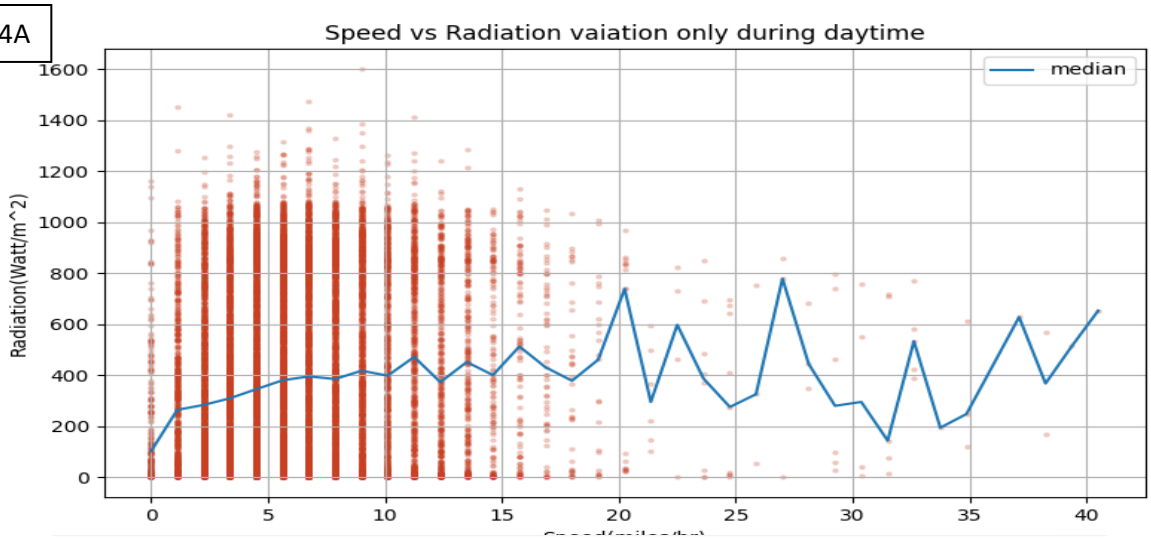
3A


Humidity vs Radiation vaiation only during daytime
median
Radiation(Watt/m^2)
Humidity(%)
0
200
400
600
800
1000
1200
1400
1600
20
40
60
80
100

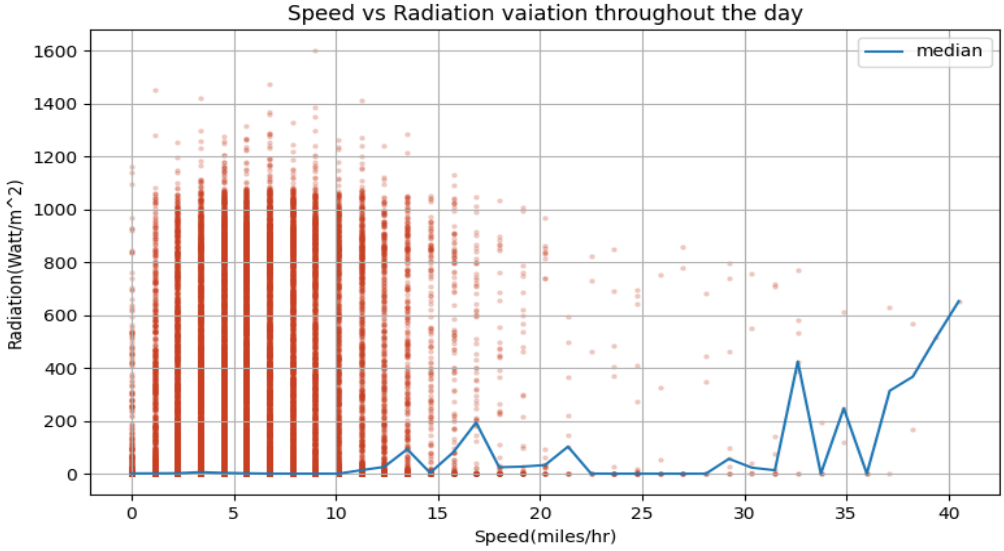
3B


Humidity vs Radiation vaiation throughout the day
median
Radiation(Watt/m^2)
Humidity(%)

4A


Speed vs Radiation vaiation only during daytime
median
Radiation(Watt/m^2)

4B


Speed vs Radiation vaiation throughout the day
median
Radiation(Watt/m^2)
Speed(miles/hr)

B. Drawing Inferences

Q) during what time of day is solar radiation generally highest?
Ans) ***12:25:00***

Q) Which environmental parameter appears to have the strongest relationship with radiation?

Temperature seems to be most directly linked with solar radiation especially if only day time values are considered (graph 2a)

Even wind speed seems to have a log relation with solar radiation but a strong conclusion cant be drawn as the data for higher windspeeds is too sparce.

Q) What patterns can be observed between radiation, temperature and humidity?
Ans)

Temperature vs Radiation:

The temperature seems highly related to the radiation observed, from the graph 2a it can be seen that as temperature increases the median radiation also increases regularly. Eventhough the median is somewhat regular the actual scatter plot is still quite spread out, this suggests the radiation is also dependant on other factors.
Also it would not be correct to assume that the high radiation is caused high temperatures the other way around is much more likely.

Humidity vs Radiation:

If we look at the all day plot (3b) no relation can be drawn. When observations from the entire day are plotted, for certain humidity values it is seen the median is almost zero, this is because these humidity values may occur more commonly at night(when solar radiation is very low) as compared to the other values. Hence the day time only plot is better way to study the affect of humidity on solar radiation.
From the first graph it can be seen that as humidity increases the solar radiation decreases almost regularly, this may because the increased amount of water vapour may block some of the radiation.
Also the scatter plot is very spread out again even more then the spread in temperature graph
Hence radiation depends on other factors, with humidity not being a majour facor

Q) Are there any unusual observations or outliers? What might explain them?
Ans)

In graph 2a there are some radiation values in the temperature range of 40F-45F which are unsually high, this shows that high temperatures are not nesseserary for higher radiation. At these ranges the radiation my reach the grounf before the atmosphere has a chance to heat up.4

When we look all day plot (2b) at around 52F-57F there is a sharp increase in radiation, this could indicate that these temperatures occur at the morning time lead to a inc in radiation as sun rises.

The median radiation increases approxly logerithmicly with wind speed upto the value of 15 mile/hr, after that the median starts to vary wildly. This may be because after 15 mile/hr the data is quite sparce

C. Solar Day Analysis

Data frame with proper date time column:

	UNIXTime	DateTime	Radiation	Temperature	Pressure	Humidity	WindDirection(Degrees)	Speed	TimeSunRise	TimeSunSet
0	1475229326	2016-09-29 23:55:26	1.21	48	30.46	59	177.39	5.62	06:13:00	18:13:00
1	1475229023	2016-09-29 23:50:23	1.21	48	30.46	58	176.78	3.37	06:13:00	18:13:00
2	1475228726	2016-09-29 23:45:26	1.23	48	30.46	57	158.75	3.37	06:13:00	18:13:00
3	1475228421	2016-09-29 23:40:21	1.21	48	30.46	60	137.71	3.37	06:13:00	18:13:00
4	1475228124	2016-09-29 23:35:24	1.17	48	30.46	62	104.95	5.62	06:13:00	18:13:00

Daily max and avg radiation for grouped observations:

Daily Avg:

2016-09-01	298.451600
2016-09-02	333.471219
2016-09-03	123.974574
2016-09-04	153.584321
2016-09-05	306.065125
	...
2016-12-27	239.910556
2016-12-28	174.910278
2016-12-29	125.018049
2016-12-30	88.367684
2016-12-31	144.288542

Daily Max:

2016-09-01	1073.71
2016-09-02	1062.01
2016-09-03	1088.09
2016-09-04	1601.26
2016-09-05	1264.07
	...
2016-12-27	870.52
2016-12-28	1157.27
2016-12-29	696.30
2016-12-30	439.43
2016-12-31	1026.34

Date with the highest recorded solar radiation: 2016-09-04

Time at max radiation and environmental conditions:

Time	12:15:04
Temperature	61
Humidity	93
Speed	9.0

Name: 6465, dtype: object

Top 5 days with highest avg radiation:

```
['2016-09-04', '2016-10-19', '2016-10-07', '2016-10-31', '2016-09-27']
```